

AUGUST/RESIT EXAMINATIONS 2022/2023

MODULE: CA216 - Operating Systems

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 2,0

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[8 Marks]**

Provide a labelled diagram of the Linux process life-cycle with explanations i.e., the states a process can be in and how it can transition between these states.

Q 1(b)**[8 Marks]**

Explain the memory layout of the C program below. In your answer, state what will be stored in each of the following memory sections when the program below is executed: *text, initialised data, uninitialised data, heap and stack*.

C program for Q1(b)

```
01: #include <stdio.h>
02: #include <stdlib.h>

03: int a;
04: int b = 2;

05: int main(int argc, char *argv[])
06: {
07:     int *v;
08:     int i;

09:     v = (int *)malloc(sizeof(int) * 3);

10:     for(i=0; i<3; i++)
11:         v[i] = i * a;

12:     return 0;
13: }
```

Q 1(c)**[5 Marks]**

What is a system call? Name and describe one system call.

Q 1(d)**[4 Marks]**

Briefly explain the purpose of the following two commands on Linux:

- a) size
- b) pstree

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[5 Marks]

Using Amdahl's law, show which of the following programs has the greatest potential speedup on 10 processors.

Program A Serial proportion = 98%	Program B Serial proportion = 2%
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Q 2(b)

[11 Marks]

Explain the difference between preemptive and non-preemptive scheduling. Describe in detail one preemptive scheduling algorithm using an example.

Q 2(c)

[5 Marks]

What is processor affinity? Briefly describe one benefit of processor affinity.

Q 2(d)

[4 Marks]

Name two different ways inter-process communication can be achieved in Linux.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[7 Marks]

A semaphore is a type of synchronisation primitive. Explain the **wait()** and **signal()** operations of semaphores. In your answer, explain how a counting semaphore differs from a binary semaphore.

Q 3(b)

[11 Marks]

Explain the concept of deadlock using an example. In your answer, describe the necessary requirements for deadlock to occur, and one strategy that can be used to avoid deadlock.

Q 3(c)

[7 Marks]

A programmer is attempting to implement a solution to the bounded-buffer/producer-consumer problem for a buffer of size **n**. They have encountered difficulty when using the implementation below. Three semaphores **mutex**, **full** and, **empty** have been initialised to **1**, **0**, and **n**, respectively.

Outline what changes you would make to the implementation below for it to operate correctly. In your answer, refer to specific line numbers.

Consumer process:	Producer Process:
01: while (true) { 02: wait(empty); 02: signal(mutex); 03: ... 04: /* remove item from buffer*/ 05: ... 06: signal(full); 06: signal(empty); 07: }	08: while (true) { 09: /* produce an item */ 10: signal(empty); 11: wait(mutex); 12: ... 13: /* add produced item to buffer */ 14: ... 15: signal(full); 16: wait(empty); 17: }

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 25]

Q 4(a)

[4 Marks]

In the context of filesystems, briefly explain the difference between hard and soft links.

Q 4(b)

[10 Marks]

Paging is a memory management scheme. Explain how paging works using an example of a 32-bit virtual address. In your answer, briefly explain how paging addresses the problem of external fragmentation.

Q 4(c)

[5 Marks]

Is the following statement true or false? Explain why or why not.
Swapping/a backing store must be enabled in order for paging to work.

Q 4(d)

[6 Marks]

Using an example, explain the benefit of acyclic-graph directory structures. Explain your answer.

[End of Question 4]

[END OF EXAM]